Intelligent Retail Risk Monitoring

Graph-Based Behavior Modeling, Adaptive Suspicion Scoring, and Explainable Alerts

Team Members

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PROBLEM STATEMENT AND PROPOSED SOLUTION

Problem Statement

- Retail stores lose significant revenue to theft each year
- CCTV feeds are still manually monitored in most settings
- Human operators miss subtle behavioral patterns, especially across multiple zones
- Existing systems rely on static rules (e.g. flag anyone over 5 min) with no behavioral context
- Detection-only systems answer 'is someone there?' but not 'what are they doing?'

Proposed Solution

- Detect and persistently track all customers using YOLOv8 and DeepSORT
- Map movements onto a semantic zone graph representing the store layout
- Extract behavioral features per person: dwell time, zone revisits, trajectory irregularity, and billing bypass
- Compute an adaptive suspicion score that calibrates to the store's own normal behavior
- Generate explainable alerts for flagging
- Display on real-time dashboard with heatmaps and risk timelines

POTENTIAL CHALLENGES

1

Occlusion and Re-identification Failures

Crowded store environments cause frequent occlusions. DeepSORT's appearance CNN may fail to re-identify a person correctly after they are blocked for several seconds, causing ID switches and broken trajectories.

2

Defining and Calibrating Zone Boundaries

Store layouts vary. Poorly defined boundaries produce incorrect zone assignment, degrading all downstream behavioral features.

3

Adaptive Score Calibration on Limited Data

The suspicion score weights need to be calibrated from observed normal behavior. In a new or low-traffic store, early estimates of 'normal' will be noisy, increasing false positives initially.

4

Computational Overhead for Real-time Operation

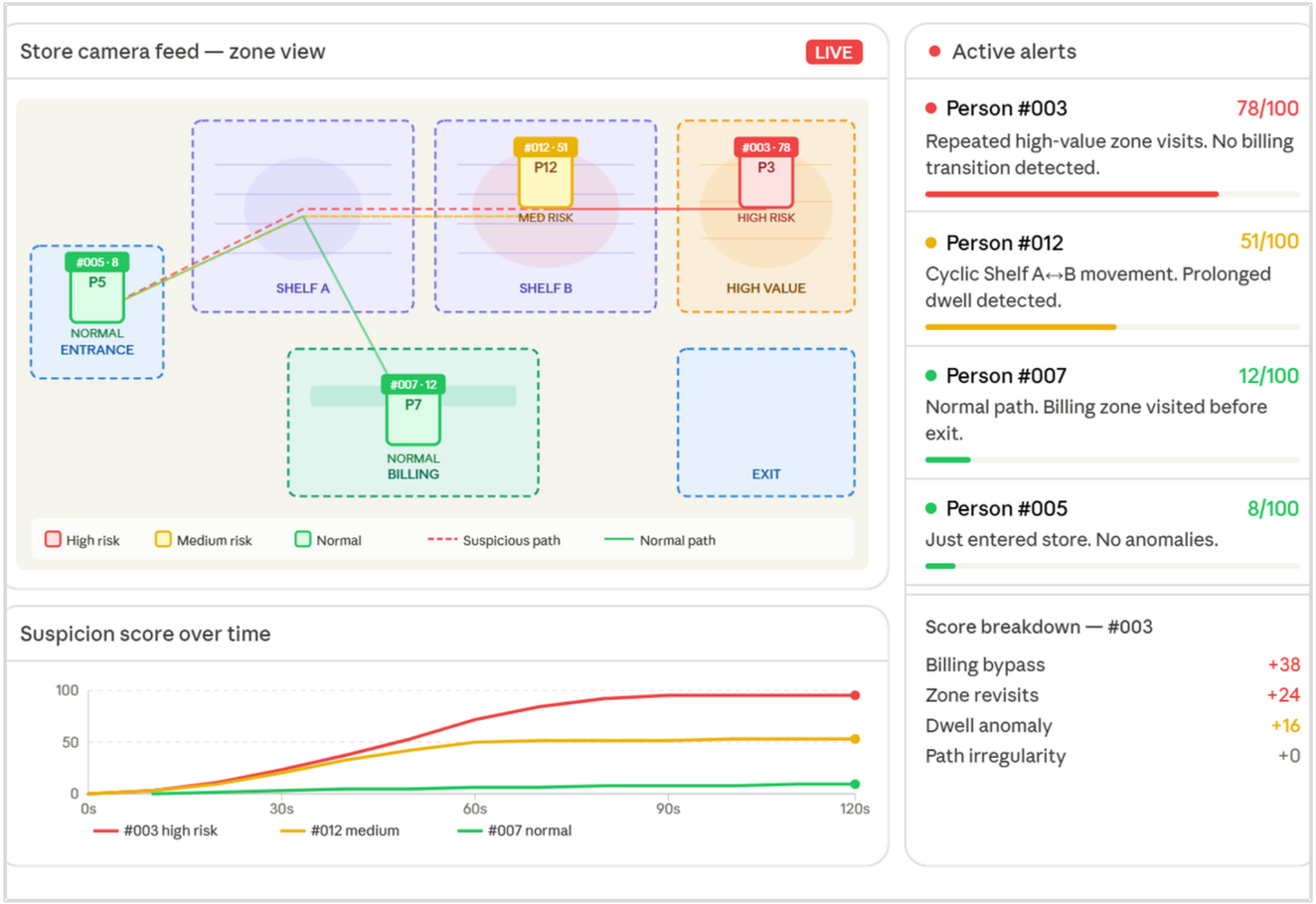
Running YOLOv8, DeepSORT, zone mapping, and the dashboard simultaneously on a single machine requires careful optimization. Frame drops or latency spikes could make the system impractical in live settings.

CODING ENVIRONMENT, LANGUAGE, AND EXPECTED OUTPUT

Environment and Stack

- Language: Python 3.10
- Detection: YOLOv8 (Ultralytics)
- Tracking: DeepSORT / ByteTrack
- Graph modeling: NetworkX
- Dashboard: OpenCV + Streamlit
- Hardware: NVIDIA GPU (CUDA)
- Dataset: Custom retail CCTV footage or publicly available surveillance datasets (e.g. MOT17, RetailS, UCF-Crime, VIRAT, etc)

Example Output (Dashboard)



FALLBACK PLAN

Uncertainty-Aware Hybrid Medical Image Diagnosis using nnU-Net and Xception

Problem and Approach

- Medical AI systems often give predictions without any confidence estimate
- Incorrect predictions without uncertainty measures are risky in clinical settings
- Segmentation (nnU-Net) and classification (Xception) are treated separately in most pipelines
- This fallback proposes a bidirectional framework integrating both tasks with uncertainty estimation
- Disagreement detection flags cases where segmentation and classification conflict

Target results if we pursue the fallback

- Segmentation metric: Dice Score > 0.80, IoU > 0.70
- Classification metric: Accuracy > 85%, ROC-AUC > 0.90
- Uncertainty measured via Monte Carlo Dropout and Expected Calibration Error
- Datasets: BraTS (brain tumor), ISIC (skin lesion), chest X-ray
- Explainability via Grad-CAM overlaid on segmentation masks
- Not a backup topic: it is a research-grade project with independent novelty

REFERENCES

[1] Wojke, N., Bewley, A., and Paulus, D. (2017). Simple Online and Realtime Tracking with a Deep Association Metric. IEEE ICIP.

Relationship to our project:

This paper introduces DeepSORT. It proposes combining a Kalman filter for motion prediction with a CNN-based appearance descriptor for re-identification across frames. Our system depends on this to build per-person trajectories, and behavioral analysis including zone mapping and suspicion scoring.

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[2] Jocher, G., et al. (2023). Ultralytics YOLOv8. GitHub. https://github.com/ultralytics/ultralytics

Relationship to our project:

YOLOv8 is the object detector powering Stage 1 of our pipeline. It performs single-pass real-time inference to localize every person in each video frame. The bounding box outputs are passed directly to DeepSORT for tracking.

Originality Disclosure

Does this project overlap with any other coursework or previous projects?

- This project does not reproduce any previously submitted work.
- No part of this proposal has been submitted to another course, past or present.
- The fallback project (medical imaging) is also independently scoped and does not overlap with prior submissions.

How is this submission substantially distinct?

- Three novel contributions combined: graph zone paths, self-calibrating score, and explainable alert decomposition
- G = (V, E) store graph encodes customer paths as node sequences, not standard in any CV course project
- Score weights calibrate to the store's own behavioral distribution, no manually defined thresholds
- XAI layer decomposes each score into: dwell anomaly, billing bypass, transition frequency, and trajectory irregularity